



DESCRIPTION OF COURSEWORK

Course Code	SWE402
Course Name	Data Mining
Lecturer	Talal Ali
Academic Session	2025/04
Assessment Title	Assignment

A. Introduction/ Situation/ Background Information

This assignment will cover basic data mining skills such as data preprocessing, exploratory data analysis, machine learning, and model evaluation. Students are expected to demonstrate their understanding of data mining course.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

- CLO 1 Describe the concept and principles of data mining and its algorithms.
- CLO 2 Apply the knowledge to design a data mining algorithm.
- CLO 3 Assess the performance of data mining algorithms for a given problem.
- CLO 4 Build programming solutions using data mining techniques for a given problem.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
 - a. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
 - b. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

- c. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This assignment is **individual-based**. You must use Python to address all the questions and submit the solution codes to Moodle before **20 Jun 2025, 11:59 AM**. The submission should include the following files:

- 1) A solution file (Jupyter Notebook/ Python source file) that includes the answers to all questions. Make sure to:
 - Name variables meaningfully.
 - Include comments in your code.
- 2) A report file (PDF file) that explains each task (no more than 30 pages). Organize the report as follows:
 - Assignment Cover Page along with the marking rubric.
 - Task Question: State the question.
 - Answer: Provide the solution code and output.
 - Explanation: Explain the logic, approach, and any assumptions made.
 - Name the files as <StudentID_Assign2>
 - Please submit **a printed hard copy** of the report to me no later than Jun 20, 2025, by 11:59 AM.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Jupyter Solution File	70
2.	Report	30
	TOTAL	100

F. Task(s)

Section A: Basic Python

1. Write a Python script to solve each of the following problems:

a. Given the variables:

```
Country = "Malaysia"
```

```
Population= 33000000
```

Use the print function to display the following string based on the format below.

```
The population of Malaysia is 33,000,000.
```

[1 Marks]

b. Given a list of integers:

```
numbers = [12, 15, 18, 21, 24, 27, 30, 33, 36, 39]
```

- Write a Python Script to print the min, max, sum of the list.
- Write a Python Script to print [15, 18, 21] from the list.
- Write a Python script to calculate the sum of all even numbers in the list.

[2 Marks]

c. Given the following nested dictionary:

```
1 data = {  
2     'key1': [1, 2, 3,  
3         {'nested_key': ['a', 'b', 'c',  
4             {'target': [10, 20, 30, 'hello']}]]]  
5 }
```

- Use indexing to grab nested_key nested dictionary.
- Use indexing to grab the word “hello.”
- Change the word “hello” to 40.

[2 Marks]

d. Write a Python function that takes a string as input and returns the number of vowels (a, e, i, o, u) in the string. Test your function with the following strings:

- "Artificial Intelligence"
- "Data Management"

[2 Marks]

e. Using the NumPy module, solve the following tasks:

- Create an array of 10 zeros.
- Create an array of all odd numbers between 10 and 50.
- Create a 3x3 identity matrix.

[2 Marks]

Section B: Data Analysis

1. You are given a dataset named **student_data.csv**. This dataset contains information about students, including their names, ages, grades, and subjects. Write a Python script to perform the following tasks:

- a. Import Pandas into your Python program and store the student data in a Pandas DataFrame. Display the first five rows of records.

[1 Marks]

- b. Find and display the number of missing values in each column. Use different ways to handle missing values.

[3 Marks]

- c. Discover and show the following information:
 - i. The average age of students.
 - ii. The name of the student with the highest grade.
 - iii. The number of students who are older than 20.
 - iv. The percentage of students who have a grade above 3.8.
 - v. The most common subject among students.
 - vi. Who is the youngest student?
 - vii. What is the average grade per subject?
 - viii. How many students are enrolled in each subject?
 - ix. Who are the students with grades below 2.5?
 - x. What is the median grade of students?

[10 Marks]

Section C: Data Preprocessing

Using the **Titanic** dataset from Seaborn, answer the following questions.

1. Data Preprocessing and Exploration:

- f. Import the necessary Python libraries and load the Titanic dataset from Seaborn.
- g. Display the first five rows of the dataset and the last ten rows.
- h. Display a summary of the dataset, including column types and missing values.
- i. Handle missing values using at least three different ways.
- j. Convert categorical variables into numerical representations where applicable.
- k. Perform feature selection to identify relevant columns for analysis and machine learning.

[7 Marks]

2. Data Visualization

Use Seaborn and Matplotlib to explore patterns in the **titanic** dataset from seaborn (use the pre-processed dataset from section C). Interpret insights from the visualizations.

- a. Visualize the distribution of passengers' genders (sex)
- b. Show the distribution of passengers' ages (age).
- c. Display the number of passengers in different classes (pclass).
- d. Show the distribution of fares paid (fare).
- e. Compare age distributions across different passenger classes (pclass).
- f. Show survival status grouped by gender (sex).
- g. Show survival status grouped by passenger class (pclass).
- h. Compare the age distribution of survivors and non-survivors.

[16 Marks]

Section D: Machine Learning & Model Evaluation

1. Supervised Learning - Passenger Survival Classification (use the pre-processed dataset from section C):

Train a classification model to predict whether a passenger survived (survived column).

- a. Split the dataset into training (80%) and testing (20%) sets.
- b. Choose two machine learning algorithms to train a classification model.
- c. Evaluate the model using the confusion matrix.
- d. Compare the two models in terms of performance and discuss their limitations.

2. Unsupervised Learning - Passenger Clustering (use the pre-processed dataset from section C).

Apply K-Means clustering to group passengers based on age, fare, and passenger class (pclass).

- a. Select age, fare, and class as clustering features.
- b. Scale the data for better clustering performance.
- c. Apply K-Means clustering
 - Experiment with different values of k (number of clusters).
 - Use the Elbow Method to determine the optimal k.

- d. Visualize the clusters using a scatter plot.

[24 Marks]

APPENDIX 1

MARKING RUBRICS

Component Title	Assignment					Total Mark	100
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (9-8)	Good (7-5)	Average (4-3)	Need Improvement (2)	Poor (0-1)		
Section A	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	9	
	Excellent (14-11)	Good (10-8)	Average (7-5)	Need Improvement (4-3)	Poor (2-0)		
Section B	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	14	
	Excellent (23-19)	Good (18-14)	Average (13-8)	Need Improvement (7-4)	Poor (0-3)		
Section C	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	23	
	Excellent (24-19)	Good (18-14)	Average (13-8)	Need Improvement (7-4)	Poor (0-3)		
Section D	The source code fully addresses the question without compilation error. It has sensible variables.	The source code addresses the question without compilation error.	The source code partially addresses the question, but the program still works.	The source code partially addresses the question and has a compilation error.	The source code doesn't work and doesn't address the question. Or No Attempt	24	
					Subtotal	70	

	Excellent (30-24)	Good (23-18)	Average (17-14)	Need Improvement (13-8)	Poor (7-0)		
Report	The report meets all criteria exceptionally well.	The report meets most criteria well but has minor issues.	The report meets some criteria but has noticeable issues.	The report meets few criteria and has significant issues.	The report does not meet the criteria or is incomplete. Or No Attempt	30	
					Subtotal	30	
TOTAL						100	

Note to students: Please include the marking rubric when submitting your coursework.